

Ankur Desai

(248) 657 – 3805 | ardusa05@gmail.com | www.ardusa.dev | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, enabling real-time attendance tracking and cutting processing time by 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with 90% coverage gates, 20+ tests, and documentation for test reports and traceability.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Implemented GitLab CI/CD pipeline across the product development lifecycle (build, test, containerized deploy); 85% coverage across 35+ tests and documentation for V&V traceability.
- Introduced Neo4j to offload recursive relational queries from PostgreSQL, cutting p95 latency 48% (850ms → 440ms) and eliminating SQL join overhead for multi-hop traversals.
- Refactored Cypher with relationship-type indexing; traversals ran 3× faster than equivalent SQL self-joins.

PROJECTS

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel where full-body movements cast spells, block attacks, and trigger abilities against an AI opponent.
- Built real-time gesture pipeline using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference to sustain 60 FPS with sub-20ms input lag.
- Developed custom 2D vector physics engine with AABB collision detection to accurately translate live 3D skeletal coordinates to game state, maintaining <5px drift at 60 FPS.

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery*

May 2025 – Nov. 2025

- Engineered multi-client agentic voice assistant that batches natural dialogue, resolves implicit speaker intent, and dispatches real tasks across concurrent voice streams.
- Reduced RESTful API response time by 73% by offloading RAG pipeline to async Redis/Celery task queue.
- Built bidirectional Electron + Next.js client with embedded HTTP server for OAuth; real-time events via IPC and webhook server deliver sub-50ms propagation.

TECHNICAL SKILLS

Languages C/C++, Python, Java, JavaScript/TypeScript, SQL, Cypher, Bash
Systems & Testing Real-time systems, software testing, documentation, V&V
Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis
Databases PostgreSQL, Neo4j, MongoDB
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitLab CI/CD, GitHub Actions